ECON 100B Spring 2026 Your name and SID:

Problem Set 1 ANSWERS

Due on Gradescope: check the course Drive calendar
Consult the Late Policy in the syllabus

Instructions: Work with this file as a template, perhaps by File: Download: Microsoft Word. Other options, like handwriting, are fine as long as they are legible. Name and SID at the top.

Did you work with other students? Identify them below. **If you work with others, you should write answers in your own words.** (If you understand your answers, this should be easy.)

Type your name to sign the Honor Code. Save as a PDF, upload to Gradescope, and click through to locate each question part before clicking Submit. Profit.

You will see point totals associated with each question. But we will score your problem sets based on completion, not correctness. The point totals are there to make you familiar with how the exams will work, which are scored on correctness.

Questions below:

Did you work with other students? List them below:

Please type your name as an affirmation of the Honor Code at the University of California, Berkeley.

> "As a member of the UC Berkeley community, I act with honesty, integrity, and respect for others."

Type your name:

1. [14 points total] Short answers with capital answers.

(a) [2 points] Recall that GDP measured on the expenditure side is given by

$$Y = C + I + G + NX \tag{1.1}$$

Roughly how big are each of these components as a *percentage* of GDP in the U.S. today?

This shows up in the slide deck for Class 02 in ECON 100B. You could also probably Google for it or ask AI.

The consumption share is about	70%
The investment share is about	15%
The government share is about	20%
The net export share is about	–5%

For full credit on an exam, you need to get in the ballpark of the right answers and make sure they sum to 100%. Notice how the sign of net exports is important in reaching that 100%.

(b) [2 points] What exactly does investment spending create? Explain.

Investment spending creates **new capital**. We measure investment spending distinctly because unlike consumption spending, investment enhances the stock of productive capital goods that industries combine with labor and technology to produce output of goods and services.

(c) [2 points] In the first half of ECON 100B, what is capital? Describe it in a few sentences. Maybe describe the economic meanings of the word that do NOT directly apply to our study of economic growth.

In the first half of ECON 100B when we study economic growth, **capital** means physical stuff that produces other stuff. Capital includes houses, office buildings, construction equipment and other durable equipment, dentists' chairs, computers, and so on. These all are physical assets used in production of the goods and services that make up GDP.

The economic meaning of **capital** that is less relevant in the first half is **value** or **financial assets,** usually denominated in currency and in many cases readily transformable into cash. (If you take a course on monetary or financial economics or economic history, it is also true that we can classify financial assets by how liquid they are, and traditionally the Fed focused on a highly liquid monetary aggregate called M1.) While there is an obvious link between this definition of capital and the physical assets used in production, this second definition is more fluid (pun intended) and less precise for what we need.

(d) [2 points] Recall that real GDP Y^R equals nominal GDP Y^N divided by the price level:

$$Y^R = \frac{Y^N}{P} \tag{1.2}$$

Take the growth rate of both sides of equation (1.2), and recall that the growth rate of the price level is inflation and denoted by a lowercase Greek letter "pi," π. If the growth rate in nominal GDP is 5% and inflation is 2%, what is the growth rate of real GDP?

Taking growth rates of both sides of the equation, we find:

$$g\left[Y^R\right] = g\left[\frac{Y^N}{P}\right] = g\left[Y^N\right] - g[P] = g\left[Y^N\right] - \pi = 0.05 - 0.02 = 0.03$$

and thus the answer is that real GDP is growing by 3%.

(e) [2 points] In Chapter 4, we will see that GDP is modeled well by this Cobb-Douglas production function:

$$Y = A\, K^{\alpha}\, L^{1-\alpha} \tag{1.3}$$

where α is a constant number between 0 and 1 that is also capital's share of income. In the textbook, $\alpha = 1/3$, which is an appropriate level of the parameter for U.S. data.

Take the growth rate of both sides of equation (1.3), treating A, K, and L as variables that potentially have growth rates. Simplify and briefly describe your answer:

After taking growth rates of both sides of equation (1.3), we find:

$$g[Y] = g\left[A\, K^{\alpha}\, L^{1-\alpha}\right] = g[A] + g\left[K^{\alpha}\right] + g\left[L^{1-\alpha}\right] = g[A] + \alpha\, g[K] + (1 - \alpha)\, g[L]$$

In words, the growth rate of GDP equals the growth rate of A plus α times the growth rate of K plus $1 - \alpha$ times the growth rate of L.

(f) [2 points] Using your answer above, describe what happens to Y if A is constant while K and L both increase by 2%. Be specific and briefly explain.

If A is constant, then its growth rate is zero. Then the growth rate of Y is just

$$g[Y] = \alpha\, g[K] + (1 - \alpha)\, g[L] = \alpha \times 0.02 + (1 - \alpha) \times 0.02 = 0.02$$

GDP also grows by 2%.

(g) [2 points] Using your answer above, describe what happens to Y if A grows by 3% while K and L both increase by 2%. Be specific and briefly explain.

If $g[A] = 0.03$, then we add its growth rate into the growth rate of Y, which becomes

$$g[Y] = g[A] + \alpha\, g[K] + (1-\alpha)\, g[L] = 0.03 + \alpha \times 0.02 + (1-\alpha) \times 0.02 = 0.05$$

GDP also grows by 5%.

2. [12 points total] Cobb-Douglas Come Out to Play?

Although there are some phenomena in economics that the Cobb-Douglas function cannot capture realistically, it is a remarkably versatile tool that you will likely see again and again in your undergraduate studies. Recall that in Chapter 4, we see that the U.S. macroeconomy can be modeled well using Cobb-Douglas production:

$$Y = A\,K^{\alpha}\,L^{1-\alpha} \qquad (2.1)$$

where $\alpha = 1/3$ is capital's share of income. When we take first partial derivatives of equation (2.1) we reveal the useful functional forms of the marginal products of capital (MPK) and of labor (MPL), which in competitive markets equal their factor prices: the interest rate r and the wage w:

$$\frac{\partial Y}{\partial K} = MPK = r \qquad r = \frac{1}{3}A\left(\frac{L}{K}\right)^{2/3} \qquad (2.2)$$

$$\frac{\partial Y}{\partial L} = MPL = w \qquad w = \frac{2}{3}A\left(\frac{K}{L}\right)^{1/3} \qquad (2.3)$$

(a) [2 points] Take the growth rate of both sides of equation (2.2), which shows r as a function of A, K, and L, and simplify. If A is constant, then what must be true about the growth rate of r when labor and capital are growing at the same rate and thus the capital-labor ratio is constant? Show your work.

Taking growth rates of both sides of equation (2.2) produces

$$g[r] = g\left[\frac{1}{3}A\left(\frac{L}{K}\right)^{2/3}\right] = g\left[\frac{1}{3}\right] + g[A] + g\left[\left(\frac{L}{K}\right)^{2/3}\right] = g[A] + \frac{2}{3}g\left[\frac{L}{K}\right]$$

$$g[r] = g[A] + \frac{2}{3}(g[L] - g[K])$$

When $g[A] = 0$ and if $g[L] = g[K]$, these terms all become zero, and $g[r] = 0$

(b) [2 points] Using your answer above, what happens to r if there is no growth in A but K rises above L by 1%?

Our previous answer shows

$$g[r] = g[A] + \frac{2}{3}(g[L] - g[K])$$

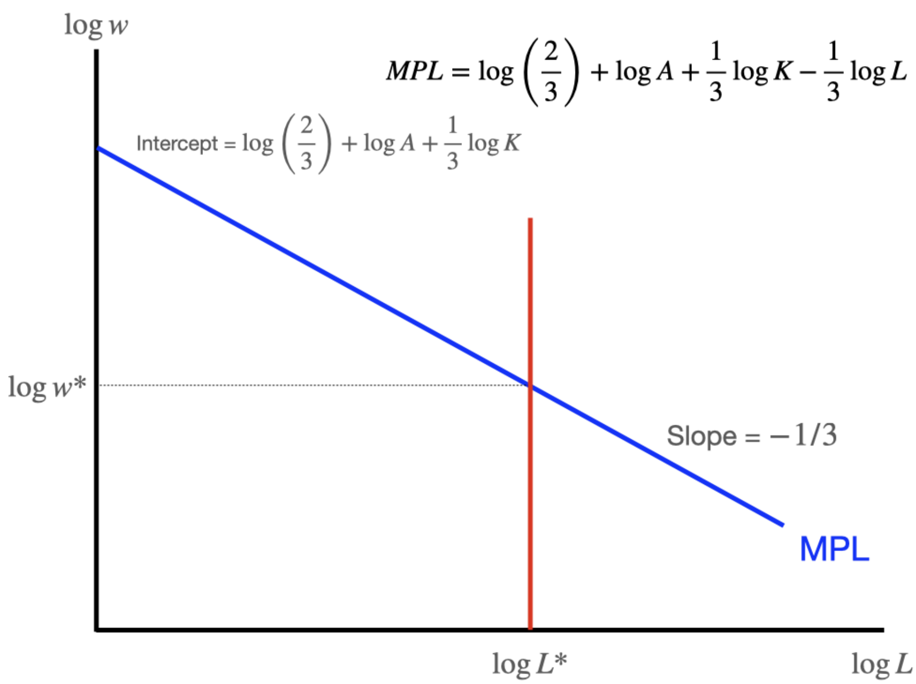
When $g[A] = 0$ and K rises 1% above L, then we can treat this as $g[L] = 0$ and $g[K] = 0.01$, which produces $g[r] = \frac{2}{3} \times (-0.01) \approx -0.0067$ or –2/3%, a reduction of two thirds of a percent.

(Minutia for those concerned: this is a percent change in r, not a percentage-point change.)

Consider what happens when we take the natural log of both sides of equation (2.3), which is similar to taking the growth rate:

$$\log w = \log \frac{2}{3} + \log A + \frac{1}{3} \log K - \frac{1}{3} \log L \qquad (2.4)$$

(c) [3 points] In the **labor market**, assume technology A and capital K are both fixed. On the axes below, graph the simple relationship that you see in equation (2.4) between the log wage on the vertical axis and $\log L$ on the horizontal axis. Explain what you drew. Report the numerical values of the vertical intercept and the slope.

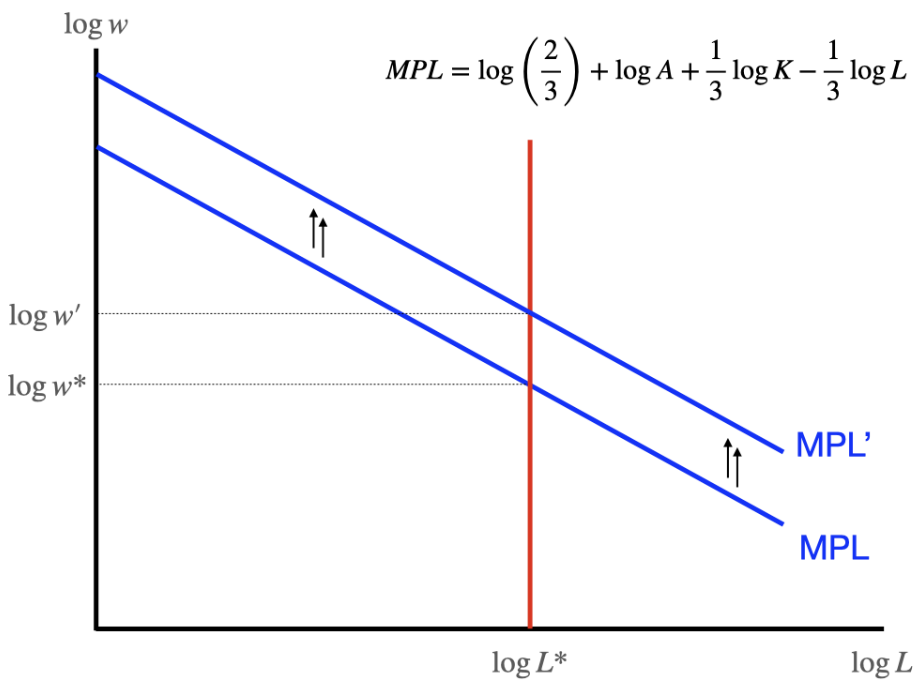


The log marginal product of labor here is a downward sloping straight line with slope equal to –⅓ and vertical intercept equal to $\log \frac{2}{3} + \log A + \frac{1}{3} \log K$, basically everything in the math except for the horizontal variable, $\log L$.

(d) [2 points] In the picture you just drew, add a vertical **labor supply curve** and show the equilibrium wage and employment in the labor market. Briefly explain below.

We depict labor supply as perfectly inelastic, a vertical line at $\log L^*$ that crosses the labor demand curve at $\log w^*$.

(e) [3 points] Redraw the labor market on the axes below and depict and describe what would happen in the labor market if the amount of **capital** in the economy were to increase, other things equal.



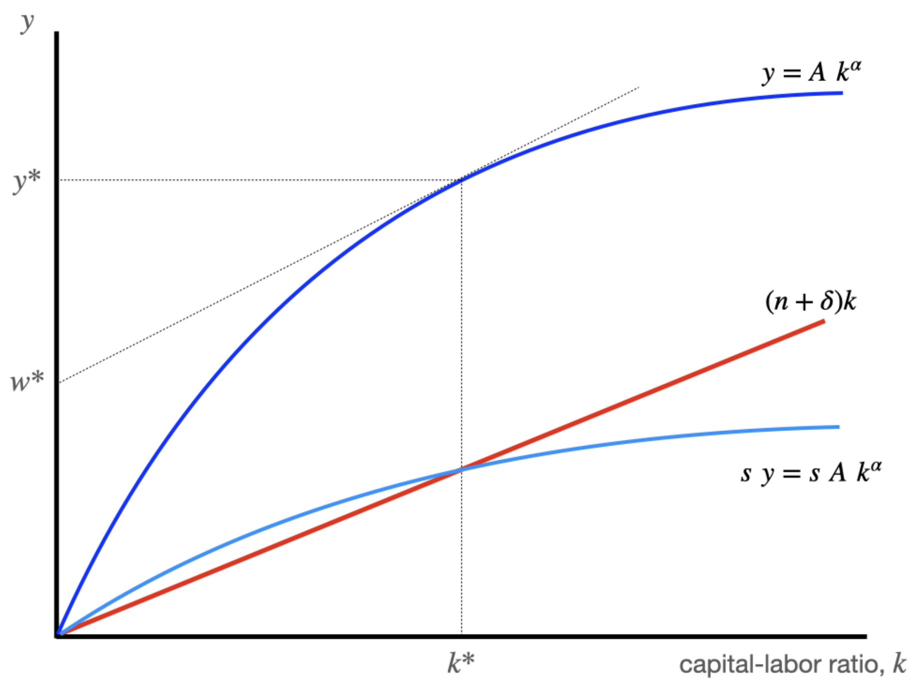
If capital increases, then log capital does as well, producing an increase in the intercept and an upward (or outward or rightward) shift in labor demand. With labor unchanged, more capital makes workers more productive, and the market-clearing wage will rise.

3. [16 points total] So how low can we go, Solow? A capital response to immigration?

In the standard neoclassical growth model of Robert Solow without technological growth, consumers supply labor inelastically and save out of income at rate s. There are constant returns to scale in labor and capital, and productivity A is fixed and not changing. Capital depreciates at rate δ, and labor increases at the rate of population growth, n. When output is Cobb-Douglas,

capital per worker is	$k = K/L$	also called the "capital-labor ratio"
output per worker is	$y = A\,k^{\alpha}$	and typically $\alpha = 1/3$
saving per worker is	$s\,y = s\,A\,k^{\alpha}$	
depreciation per worker is	$(n + \delta)k$	

The Solow diagram below shows how capital per worker reaches a *steady state* at k*, where depreciation per worker equals saving per worker: $(n + \delta)k = s\,y$, so capital per worker is unchanging there. The steady state is shown by the vertical dashed line in the diagram below.

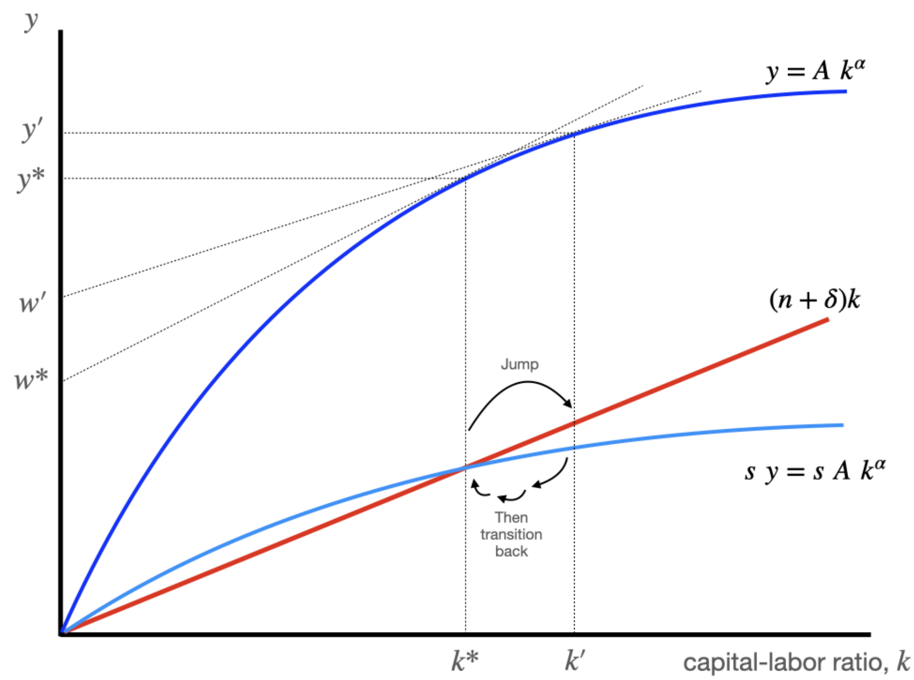


(a) [2 points] If the capital per worker is in steady state at k* and not changing over time, what is true about the wage w*, which is shown in the graph on the y-axis?

The wage w* **is also** in steady state and **not changing over time.** The wage is shown by extending the tangent line to the output per worker function back to the y-axis. (Intuitively, the wage is the marginal product of labor, which remains the same if capital per worker is fixed.

Suppose that the economy is in steady state, and then the government carries out a large removal of immigrant workers, one that increases capital per worker k by about 25%. The policy removes workers but leaves capital K and productivity, A, unchanged.

(b) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n, δ, s, A, and α change?]



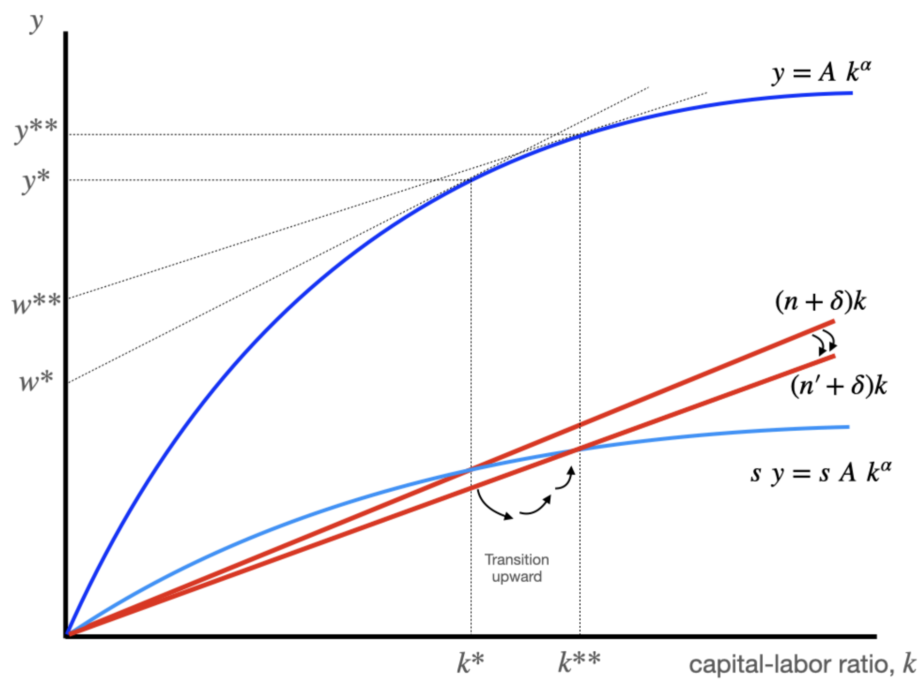
When labor is quickly removed, capital per worker k jumps up from k* to k', and the wage jumps from w* to w' in the short run.
But in the long run, capital per worker falls back from k' to k* because saving per worker (the light blue line) is less than depreciation per worker (the red line).

(c) [2 points] A higher level of capital per worker will produce a higher wage, because workers are more productive. And removing any type of worker will raise capital per worker. But is that change temporary or permanent? Why?

The change is **temporary** because over time, capital depreciates or breaks, and capital can also be "diluted" by spreading it across more workers if the population growth rate n is positive. Capital does not reproduce itself; it must be created by investment = saving. Without a rise in the saving rate, capital per worker must return to its original steady state.

Now suppose that while in steady state, the government instead decides to change **immigration policy**, reducing number of refugee and work visas each year, which reduces the rate of population growth, n, to a new level $n' < n$. (Assume there is no removal policy.)

(d) [4 points] Illustrate this policy by augmenting the graph from part (a) as needed. Draw any new curves and show jumps and changes in variables. What happens to the wage in the short run? What happens to the wage in the long run? Explain fully. [Hint: do any of the parameters like n, δ, s, A, and α change?]



When the population growth rate n falls, capital per worker remains at k^* in the short run, but the slope of the red depreciation line falls, so the line pivots clockwise. That produces saving in excess of depreciation, which increases k* eventually all the way to a new steady state k**. The **wage does nothing at first** but eventually **rises from w* to w** in the long run**.

(e) [4 points] What would have happened in the graph in pa rt (d) if immigration enhanced productivity, A, so that reducing immigration reduced the level of productivity? Explain.

If the reduction in the rate of immigration also reduced productivity relative to what it would have been, then the increase in the wage that we saw in part (d) could be partially, fully, or more than fully offset. Visually, the blue curves would both drop lower if A falls, and the steady-state k** would lie somewhere to the left, possibly even further left than k*. If immigration feeds productivity, reducing the rate of immigration might also reduce the wage.